4.Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

Aim:

To write program for continuous Charge Distribution in electric field. The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using MATLAB.

Test case (i)

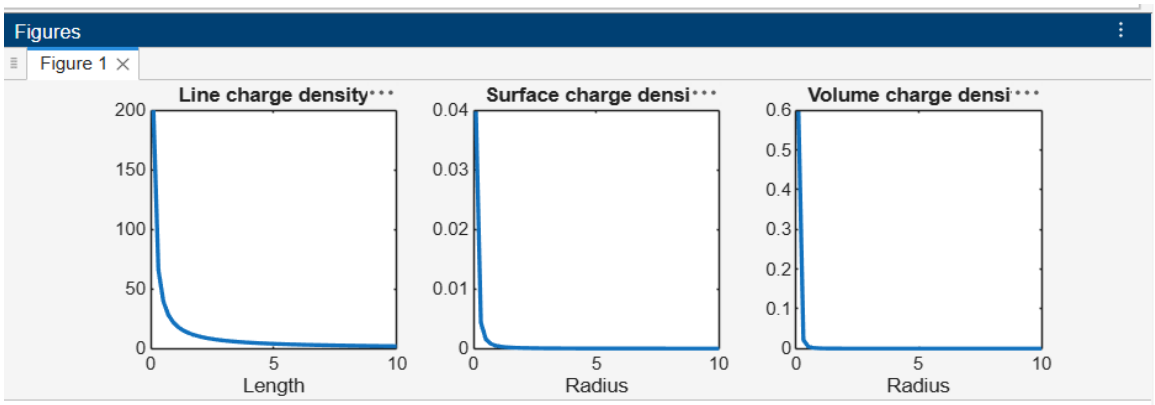
Determination of line, surface and volume charge density for given charge, length and radius

PROGRAM:

```
clc;
clear;
q=input("Enter the charge value is:");
l=input("Enter the length value is:");
% Line charge density
rho_l = q/l;
disp('the line charge density is');
disp(rho_l);
% Surface charge density
r=input("Enter the radius value is:");
rho_s = q/(4*pi*r^2);
disp('the surface charge density is');
disp(rho_s);
% Volume charge density
rho_v = q/((4/3)*pi*r^3);
disp('the volume charge density is');
disp(rho_v);
% Plot
x = linspace(0.1,10,50);
```

```
y = rho_l./x;
z = rho_s./(x.^2);
v = rho_v./(x.^3);
figure;
subplot(1,3,1);
plot(x,y,'LineWidth',2);
title('Line charge density');
xlabel('Length');
subplot(1,3,2);
plot(x,z,'LineWidth',2);
title('Surface charge density');
xlabel('Radius');
subplot(1,3,3);
plot(x,v,'LineWidth',2);
title('Volume charge density');
xlabel('Radius');
```

OUTPUT:



Test case (ii) A long wire of length L = 10 m carries a uniform charge of Q = 50 μC. Calculate the line charge density and plot electric field variation

PROGRAM:

```
clc;

L = 10;

Q = 50e-6;

lambda = Q/L;

disp("Line charge density:");

disp(lambda);

r = linspace(0.1,5,100);

k = 9e9;

E = k*lambda./r;

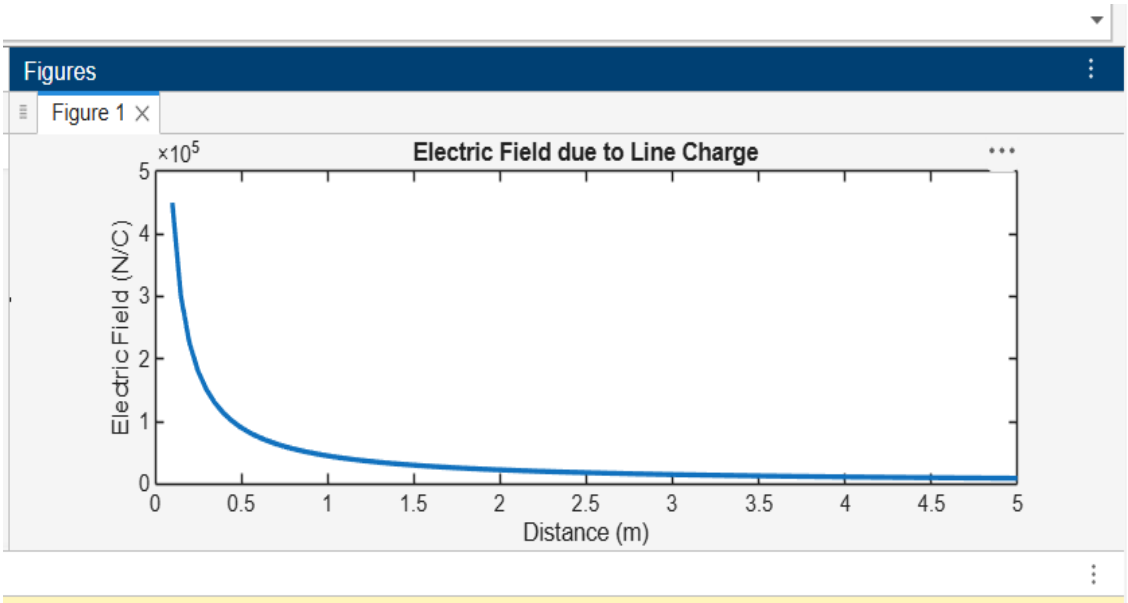
plot(r,E,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Electric Field (N/C)');

title('Electric Field due to Line Charge');
```

OUTPUT:

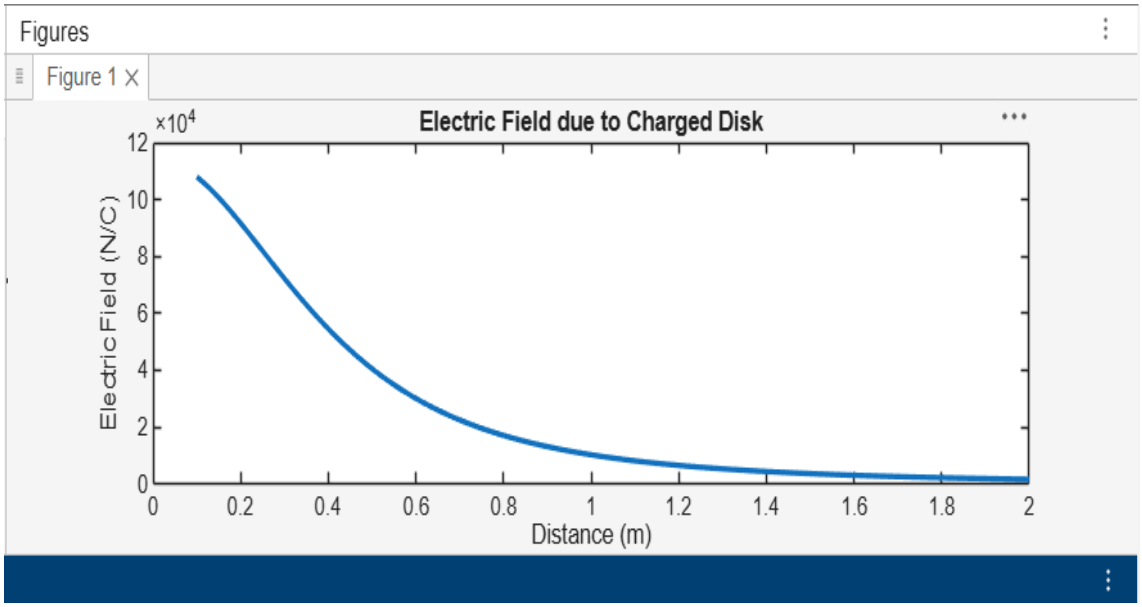


Test case (iii) A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 μC. Calculate surface charge density and electric field

PROGRAM:

```
clc;
R = 0.5;
Q = 10e-6;
A = pi*R^2;
sigma = Q/A;
disp("Surface charge density:");
disp(sigma);
z = linspace(0.1,2,100);
k = 9e9;
E = (k*sigma*R^2)./(2*(z.^2 + R^2).^(3/2));
plot(z,E,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Charged Disk');
```

OUTPUT:



Test case (iv)

A uniformly charged sphere of radius R = 2 m with total charge Q = 20μC. Calculate volume charge density and electric field inside and outside 3

PROGRAM:

```
clc;

R = 2;

Q = 20e-6;

V = (4/3)*pi*R^3;

rho = Q/V;

disp("Volume charge density:");

disp(rho);

r = linspace(0.1,3,100);

epsilon0 = 8.854e-12;

k = 1/(4*pi*epsilon0);

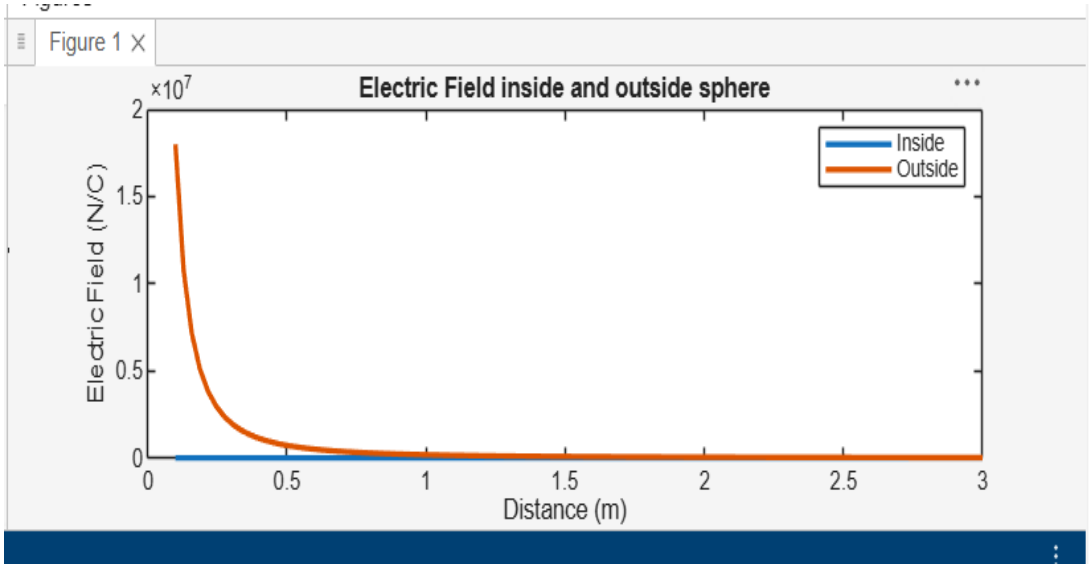
E_inside = (rho.*r)/(3*epsilon0);

E_outside = k*Q./(r.^2);

plot(r,E_inside,'LineWidth',2); hold on;
```

```
plot(r,E_outside,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field inside and outside sphere');
legend('Inside','Outside');
```

OUTPUT:



RESULTS:

The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density has been done using MATLAB.

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks given**	**Marks Obtained**
Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25	2.5	

		for both cases.						
Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculations are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	
Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omission2s.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30	3	
Explain and report results	Explain and report results	Explains attraction/ repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20	2	
Total						100	10	